SPE-229946-MS

Mesh Matters: Evaluating the Long-Term Effects of Proppant Size on Stimulation Efficiency in Low-Quality Formations

Iqbal Tri Putra, Febrian Jenetrius Naben, Azarico Putra, Ihsan Ramadhan, and Muhammad Rahim Putra Verdy, Pertamina Hulu Rokan

 DOI 10.2118/229946-MS

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

This paper evaluates the long-term impact of proppant size selection on hydraulic fracturing performance in the BSO field, where the target formations are characterized by low-quality, ductile reservoirs. Initially, 12/18 mesh proppant was used in over 40 wells, before shifting to 20/40 mesh based on expert recommendation to avoid potential proppant embedment. Following suboptimal results with 20/40 mesh, a full technical lookback was conducted to determine the optimal design direction for future development.

Hydraulic fracturing design involves key inputs, including proppant size, fluid type, pad ratio, and pumping schedule, alongside reservoir data. In this study, all parameters were held constant except proppant size. Performance comparison was conducted between ~30 wells with 20/40 mesh and ~40 wells with 12/18 mesh. Evaluation metrics included normalized cumulative production, productivity index (PI), pump run life, and proppant flowback behavior. Fracture geometry vs. productivity index (PI), was analyzed to determine the stimulation efficiency of each design. Furthermore, Pressure Transient Analysis (PTA) was performed on 5 older wells with 12/18 mesh to detect potential proppant embedment.

Wells fractured with 12/18 mesh proppant demonstrated more stable production performance and superior pump reliability compared to those with 20/40 mesh, which frequently encountered flowback issues and pump failures. PTA analysis confirmed no significant proppant embedment in the 12/18 wells, with consistent fracture lengths observed after a decade of production. The geometry analysis further revealed that 12/18 mesh designs achieved effective stimulation with shorter fractures and lower total proppant volumes, whereas 20/40 mesh required longer fractures and significantly more proppant to attain comparable productivity. These findings suggest that larger proppant provides better conductivity in ductile formations, enabling fracture geometry and material optimization without sacrificing well performance.

This study offers a data-driven validation of proppant sizing decisions in ductile, low-quality reservoirs —areas where conventional wisdom often dominates design choices. By integrating PTA diagnostics and production analytics, the paper provides a practical framework to improve fracturing efficiency and reduce material requirements in future developments.

Introduction

Field Overview

The Rokan Block is predominantly composed of mature oilfields that are currently undergoing significant production decline, particularly from their High-Quality Reservoirs (HQR). In contrast, Low-Quality Reservoirs (LQR) offer substantial untapped hydrocarbon potential that remains underexploited [1]. Among the most prominent LQRs is the TE Formation, which holds the largest volume of remaining recoverable oil within the Rokan Block. This formation is particularly significant in the BSO field, located in the northern part of the block (Fig. 1). Discovered in January 1970 and brought into production the following year, BSO is characterized by a large, elongated anticlinal structure bounded by a high-angle reverse fault.

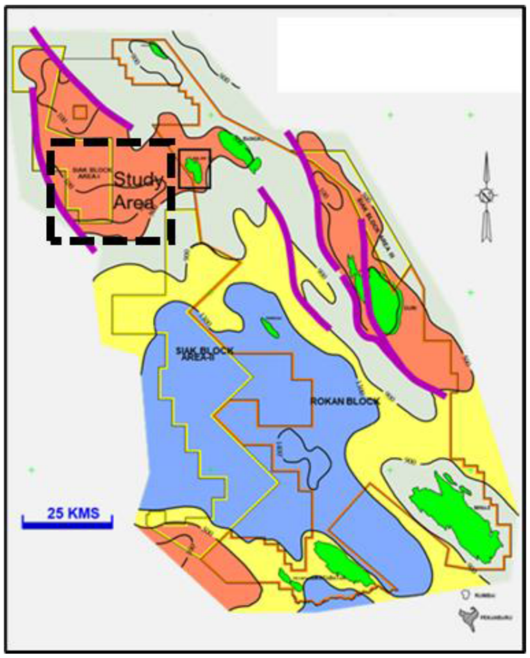


Figure 1—BSO Field

The field comprises three primary reservoirs: TE, DR, and BK. Early development activity initially prioritized the deeper, more productive DR and BK reservoirs. However, recognizing the large in-place volumes in the shallower TE Formation, a dedicated development campaign was launched in 1991. Since then, TE has contributed approximately 51 million STB in cumulative oil production and remains one of the most voluminous hydrocarbon-bearing intervals in the field. Despite its volumetric significance, the TE reservoir presents unique technical challenges, it is classified as a low-quality, ductile formation with a Young's modulus of approximately 500,000 psi and a relatively shallow depth of ~700 ft MD (Fig. 2). These mechanical characteristics, combined with a low closure pressure of around 300 psi, contribute to poor natural productivity and pose challenges for conventional stimulation [2].

As a result, the TE reservoir exhibits the lowest recovery factor among the three primary reservoirs in the field (Fig. 3), necessitating the implementation of tailored development and optimization strategies to unlock its full potential.

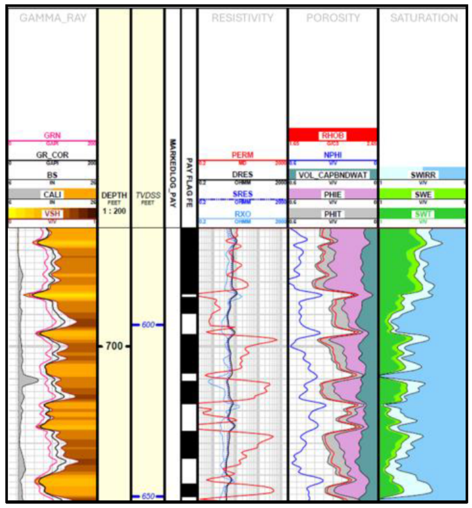


Figure 2—TE Typical Log

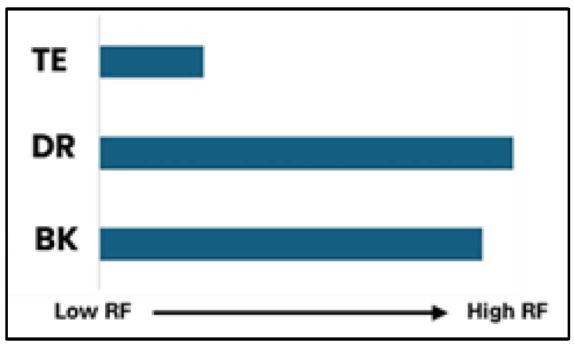


Figure 3—Three Primary Reservoir Recovery Factor

Field Development Strategies and the Emerging Challenges in TE Reservoir

To enhance production from the TE reservoir, three primary development strategies have been deployed: horizontal wells, open-hole perforated liner (OHPL) completions, and cased-hole hydraulic fracturing (Figure 4). Each of these approaches has demonstrated varying degrees of success and limitations.

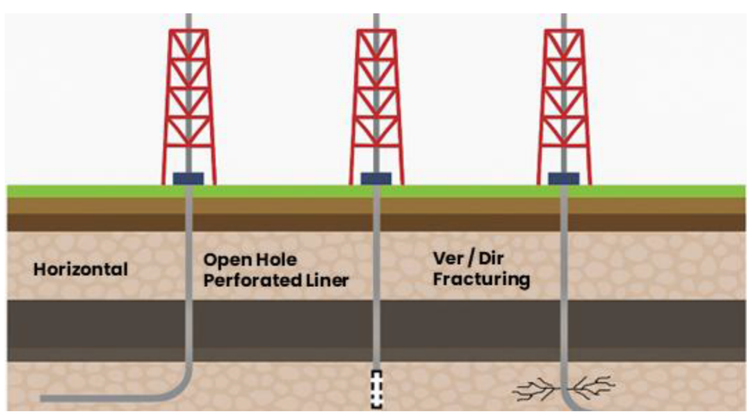


Figure 4—TE Reservoir Development Strategies

Horizontal wells were initially considered due to their potential for increased reservoir contact. However, results have been generally weak, and performance data remains limited, making it difficult to draw firm conclusions. The OHPL strategy has shown improved outcomes, particularly in wells situated near fault zones where natural fractures are more prevalent. Unfortunately, the opportunity for additional OHPL wells is constrained by limited spacing and depletion of optimal drilling locations near structural features.

As a result, cased-hole hydraulic fracturing has become a primary development method due to its flexibility in well placement [3]. This technique was applied extensively in recent years. Initially, ~40 wells were fractured using 12/18 mesh proppant, but the program later transitioned to 20/40 mesh based on expert recommendations aimed at mitigating proppant embedment risk [4].

A comparative analysis of One-year cumulative oil production per well reveals that wells completed with 12/18 mesh have outperformed those treated with 20/40 mesh (Fig. 5). The initial campaign from 2005–2008 delivered encouraging results, whereas the more recent campaign using 20/40 mesh (2022–2024) yielded notably poorer performance.

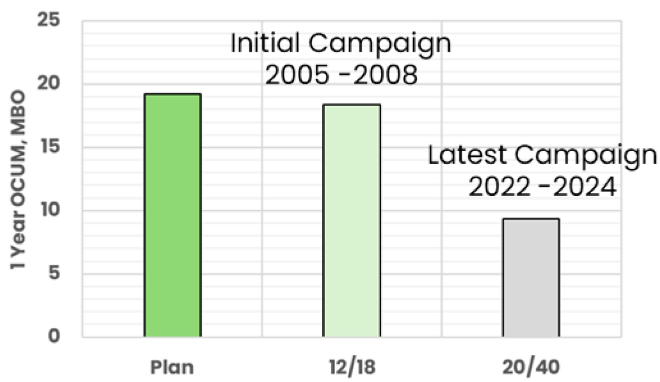


Figure 5—One Year Cumulative Oil Production Comparison

These results raise a fundamental question, what drives the performance gap between 12/18 and 20/40 mesh wells? Whether it is a function of reservoir heterogeneity, fracture effectiveness, or proppant behavior.

To resolve these uncertainties, a comprehensive technical lookback was conducted, integrating inflow and outflow analysis with supporting diagnostic tools to identify the root cause of performance differences between the mesh types, which is crucial for future field-wide development.

Methodology

Integrated Diagnostic Workflow for Fracturing Performance Review

To investigate the performance discrepancy between wells completed with different proppant mesh sizes, a comprehensive diagnostic workflow was developed, integrating both inflow and outflow analyses supported by reservoir surveillance. This approach aims to isolate the root cause of performance variance while ensuring an unbiased evaluation of all technical parameters.

The methodology is structured into three key components: inflow analysis, outflow analysis, and supporting evidence, as illustrated in Fig. 6.

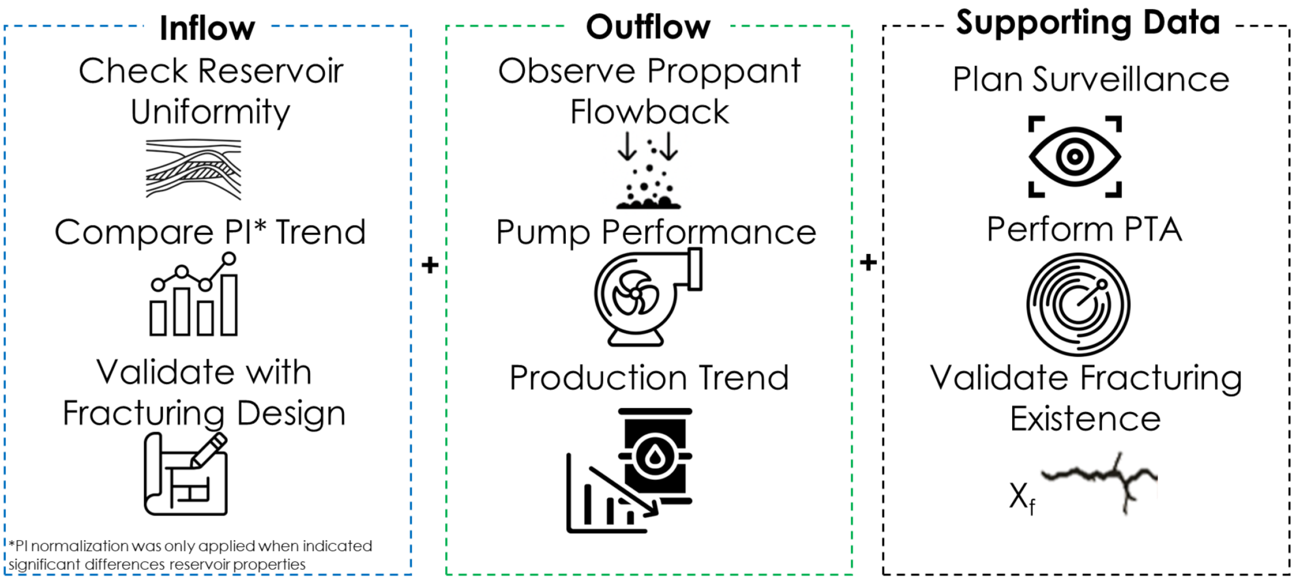


Figure 6—Integrated Diagnostic Workflow

Inflow Analysis

This stage focuses on the subsurface contribution to well productivity. It begins with validation of reservoir and rock properties across both mesh types (12/18 and 20/40), including porosity, permeability, net pay, Young's modulus, and pressure. To ensure a fair comparison, the Productivity Index (PI) was normalized only against reservoir parameters that showed significant differences between the 12/18 and 20/40 mesh groups. If no significant variation was observed, normalization was deemed unnecessary. This approach allowed the assessment of inflow performance to focus solely on stimulation-related impacts.

Subsequently, fracture stimulation design parameters were reviewed to evaluate whether one mesh type, specifically 12/18 mesh, corresponds to more effective stimulation geometry or improved inflow capacity. Key stimulation metrics such as proppant mass, fracture half-length, height, and width were used for comparison.

Outflow Analysis

The outflow component evaluates surface production reliability and its impact on well deliverability. Two sub-aspects were considered:

- Pump Performance:
  Pump run life was compared between mesh types as a proxy for artificial lift reliability and potential issues caused by sand production or mechanical stress.
- Production Performance:
  One-year cumulative oil and fluid production were analyzed across wells to determine whether different flowback or stimulation behaviors affected long-term output.

Special attention was given to identifying indicators of proppant flowback, which may manifest as shortened pump run life or abnormal decline in fluid rates.

Supporting Evidence

To substantiate the observations and eliminate ambiguity, Pressure Transient Analysis (PTA) was conducted on selected wells. PTA was chosen due to its diagnostic sensitivity in identifying flow regimes, fracture-dominated behavior, and changes in effective fracture conductivity over time.

By comparing PTA-derived parameters with post-job fracture models, the analysis aimed to validate whether any conductivity loss, such as from proppant embedment, had occurred —particularly in formations with low Young's modulus and closure pressure.

Results and Discussion

Fractured Well Distribution in BSO Field

As an initial step to ensure that reservoir placement was not a biasing factor in performance assessment, a spatial review of the fractured well population was conducted. The structural map in Figure 7 shows the distribution of wells treated with both 12/18 mesh (blue squares) and 20/40 mesh (red squares) proppants in the BSO Field. From this map, it is evident that wells completed with either mesh type are distributed across the same structural features, including high-side closures and fault blocks.

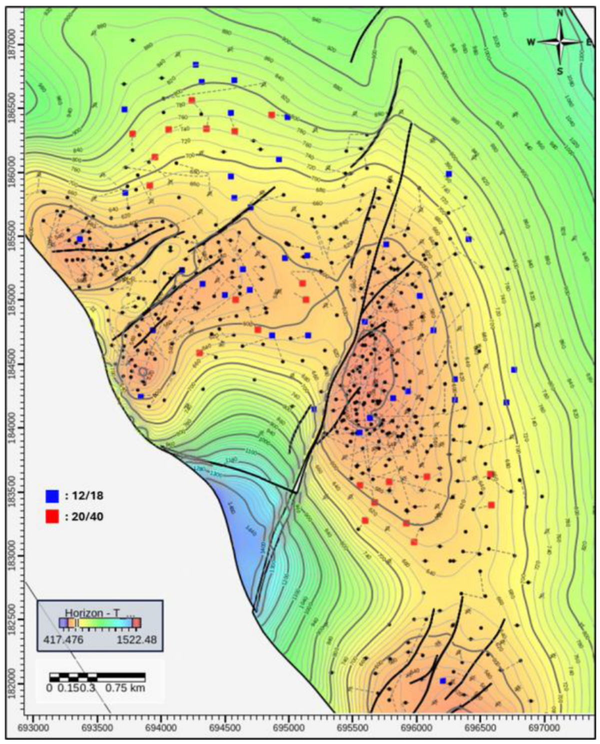


Figure 7—Fractured Well Distribution

This spatial overlap is an important validation step. It reduces the likelihood that the observed differences in well performance are attributed to geologic placement, such as favorable structure, dip, or fault proximity. Instead, it supports the hypothesis that the variance in performance is likely driven by engineering or operational parameters.

Specifically, both groups of wells are positioned within similar contour intervals and fault blocks, minimizing geological bias. This allows for a more controlled comparison, where the mesh size becomes the primary differentiating variable rather than location.

Given this spatial consistency, any significant difference in inflow or outflow behavior between the two mesh groups cannot be explained by structural positioning alone. As such, a more detailed analysis was required to investigate production mechanisms, focusing on the inflow contribution (i.e., reservoir and stimulation effects) and outflow reliability (i.e., pump performance, sand production, and cumulative recovery).

This foundational understanding of well distribution justifies the subsequent normalization of performance indicators, such as Productivity Index (PI), and the use of diagnostic data to isolate the root cause of observed performance gaps. It also reinforces the reliability of the comparative assessment framework used in this study.

Validating Reservoir Property Consistency Between Mesh Types

A foundational step in this study is the validation of reservoir property consistency between wells treated with 12/18 mesh and those with 20/40 mesh. This process is essential to isolate the impact of fracturing design and proppant size from the influence of geological heterogeneity. If significant differences in key reservoir parameters exist between the two groups, any conclusions drawn about production or inflow performance would be confounded by these underlying disparities. Therefore, a rigorous statistical assessment was performed to confirm whether both groups reside within similar reservoir conditions.

Statistical Methodology

The statistical validation was conducted using three primary tools: boxplots, histograms, and hypothesis testing via independent t-tests to calculate p-values. Each of these tools provides a unique lens through which data can be interpreted:

- Boxplots visualize the median, interquartile range (IQR), and presence of outliers across both mesh types for each property.
- Histograms offer insight into the frequency distribution and skewness of the data.
- Independent t-tests evaluate whether the means of the 12/18 and 20/40 mesh groups differ significantly for each property. The null hypothesis states that the group means are equal. A p-value less than 0.05 indicates rejection of the null hypothesis, signifying a statistically significant difference between the two groups.

This approach is consistent with engineering best practices for statistical comparison as described by Montgomery and Runger [5], who emphasized the importance of combining visual inspection with formal hypothesis testing to derive robust conclusions in comparative studies.

Reservoir Properties Analyzed

Five key reservoir parameters were evaluated for consistency:

1. Porosity
    - The porosity boxplot and histogram (Fig. 8) show broad overlaps between the 12/18 and 20/40 mesh groups, with similar central tendencies and spread.
    - The calculated p-value of 0.39 confirms no statistically significant difference.
    - This suggests both mesh groups are accessing formations of similar pore volume capacity.
2. Average Reservoir Permeability
    - Permeability data exhibits moderate dispersion, yet the shapes and ranges of the histograms are closely aligned.
    - The p-value of 0.86 strongly indicates no significant difference, supporting the interpretation that fluid transmissibility is consistent between the groups.
3. Net Pay Thickness
    - The most critical finding is that 12/18 mesh wells exhibit significantly higher Net Pay than 20/40 mesh wells.

- This is visually apparent in both boxplots and histograms and statistically confirmed by a p-value of 1.27 × 10^-5.
- The implication is that Net Pay introduces a strong bias in direct comparisons of productivity.
- As a result, it was deemed necessary to normalize the Productivity Index (PI) against Net Pay to remove this bias and enable a fair inflow performance assessment.

4. Young's Modulus
   - This mechanical property, indicative of formation brittleness or ductility, showed closely aligned distributions.
   - The p-value of 0.53 confirms no significant difference.
   - This consistency indicates that rock mechanical behavior across the two groups is broadly equivalent, and unlikely to independently influence fracturing effectiveness.
5. Pressure Gradient (Pgrad)
   - Pgrad, a parameter directly influencing fracture propagation and closure, also showed statistical similarity.
   - The p-value of 0.54 affirms no significant difference, further validating geological consistency.

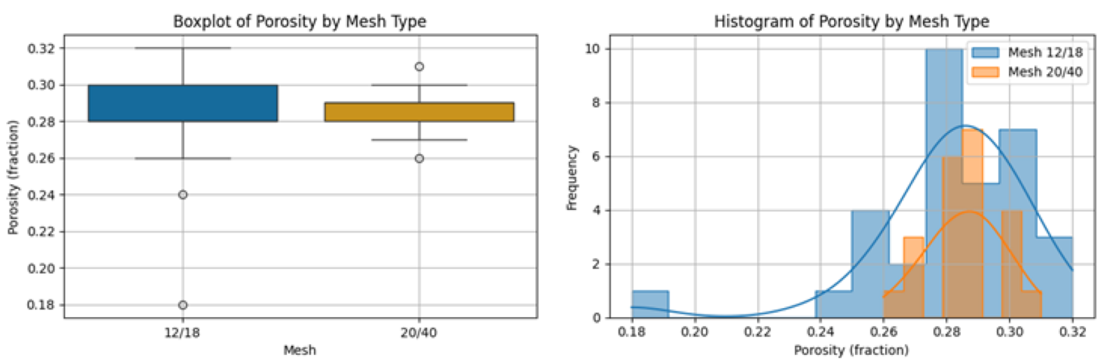


Figure 8—Box Plot and Histogram of Porosity by Mesh Type

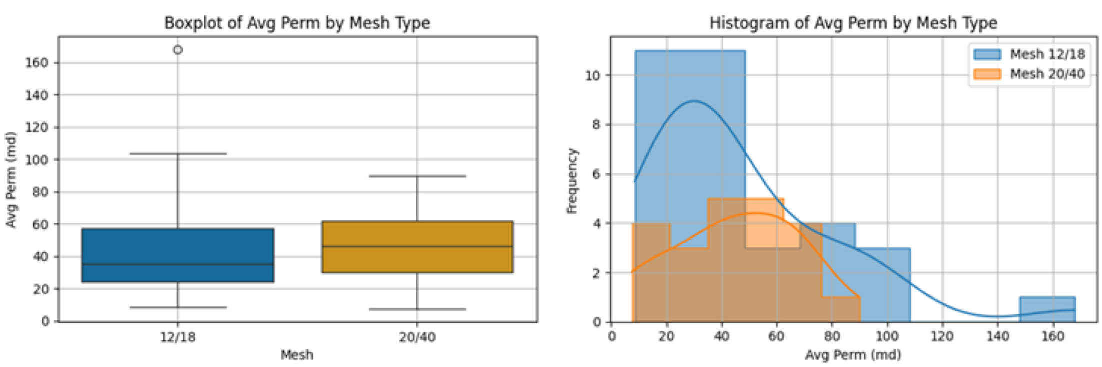


Figure 9—Box Plot and Histogram of Avg Reservoir Permeability by Mesh Type

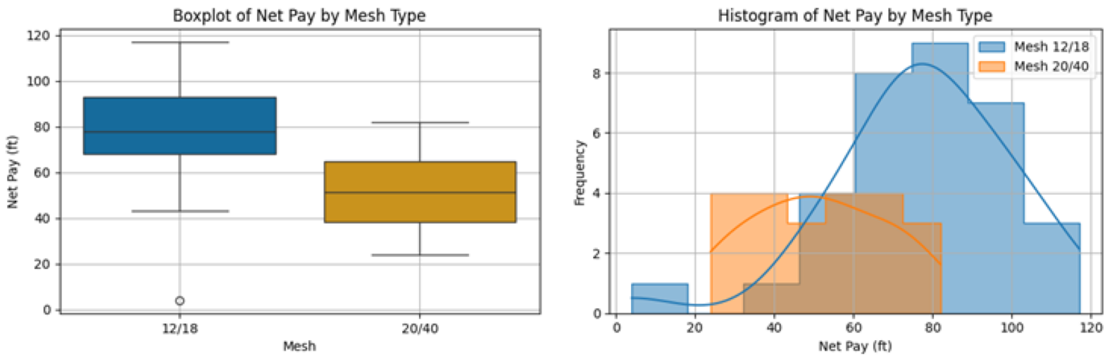


Figure 10—Box Plot and Histogram of Net Pay by Mesh Type

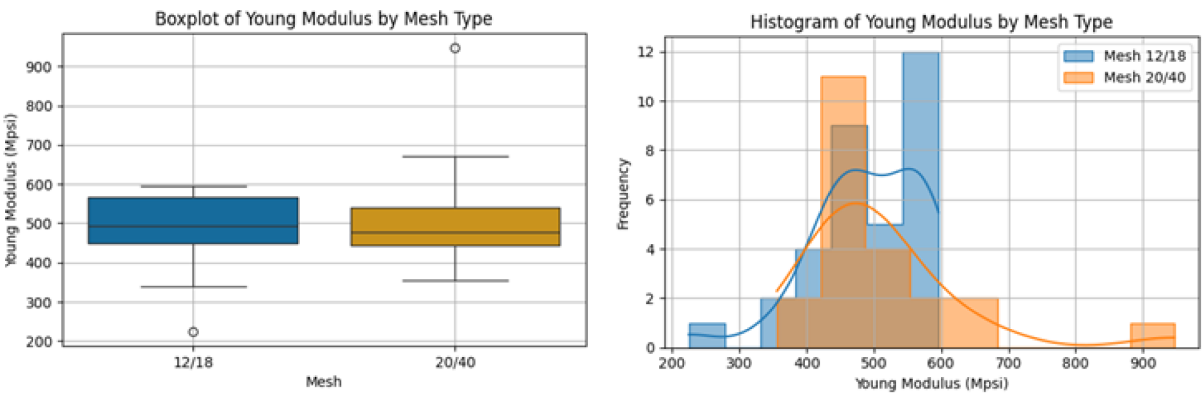


Figure 11—Box Plot and Histogram of Young Modulus by Mesh Type

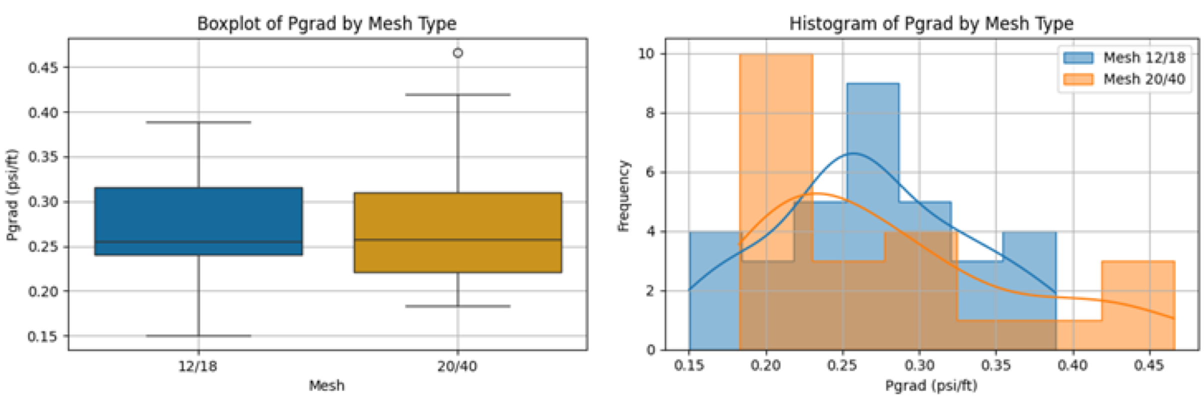


Figure 12—Box Plot and Histogram of Reservoir Pressure by Mesh Type

Interpretation and Implications

From this statistical evaluation, Net Pay is the only reservoir property exhibiting a statistically significant difference between mesh types. This variation likely reflects differences in well targeting or completion intervals rather than broader geological variability. By contrast, porosity, permeability, Young's Modulus, and pressure gradient are consistent across groups, ensuring that any observed performance differences are not attributable to these reservoir parameters.

This outcome provides the analytical justification for normalizing Productivity Index (PI) by Net Pay prior to further analysis. Such normalization aligns the inflow potential across wells, allowing a more accurate investigation into the influence of fracturing design, proppant embedment, and outflow behavior. Without this normalization step, conclusions drawn about stimulation effectiveness could be misleading, as they would conflate geological advantages with completion strategy.

In summary, this rigorous validation ensures that the comparative analysis in subsequent sections is grounded in fair and scientifically robust assumptions, laying the foundation for credible performance benchmarking between the 12/18 and 20/40 mesh completions.

Inflow and Outflow Integration Reveals Root Cause of Performance Degradation

To fully understand the disparity in production performance between wells completed with 12/18 mesh and those with 20/40 mesh proppant, this study integrates both inflow and outflow behavior alongside diagnostic indicators. This holistic approach is essential to distinguish whether observed differences stem from stimulation design, mechanical failure, or flow dynamics after the fracturing treatment.

Inflow Performance – PI Normalization

Given the statistically significant difference in Net Pay between the two mesh groups, direct comparison of Productivity Index (PI) could introduce bias. Therefore, PI was normalized against Net Pay to isolate the effects of fracture quality and stimulation efficiency from reservoir thickness [6]. The normalized PI plot demonstrates that wells completed with 12/18 mesh consistently maintain their productivity over time (Fig. 13). In contrast, 20/40 mesh wells exhibit a sharp decline in PI within the first few months after production initiation. This downward trend is a strong indicator of degradation in fractured conductivity or fluid delivery capacity, which warrants deeper investigation into potential causes.

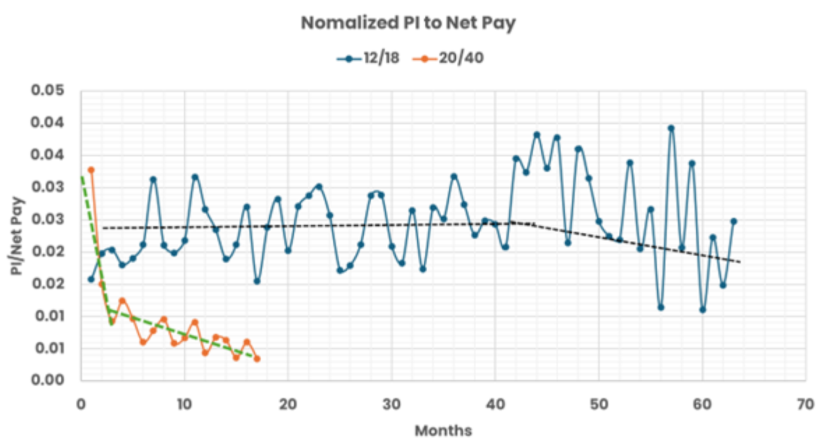


Figure 13—PI Normalization

Fracturing Geometry – Design Quality Assessment

Boxplot evaluations of stimulation parameters (fracture width, proppant volume, fracture height, and length) show that 20/40 mesh designs used significantly more proppant and resulted in wider and longer fractures (Fig. 14). Interestingly, the dimensionless fracture conductivity (Fcd) of the 20/40 mesh group, while lower, remained above the minimum best practice threshold of Fcd = 1.2. This indicates that fracture designs for both mesh sizes fell within the acceptable conductivity window, and thus, stimulation design quality alone does not explain the rapid performance decline seen in 20/40 wells.

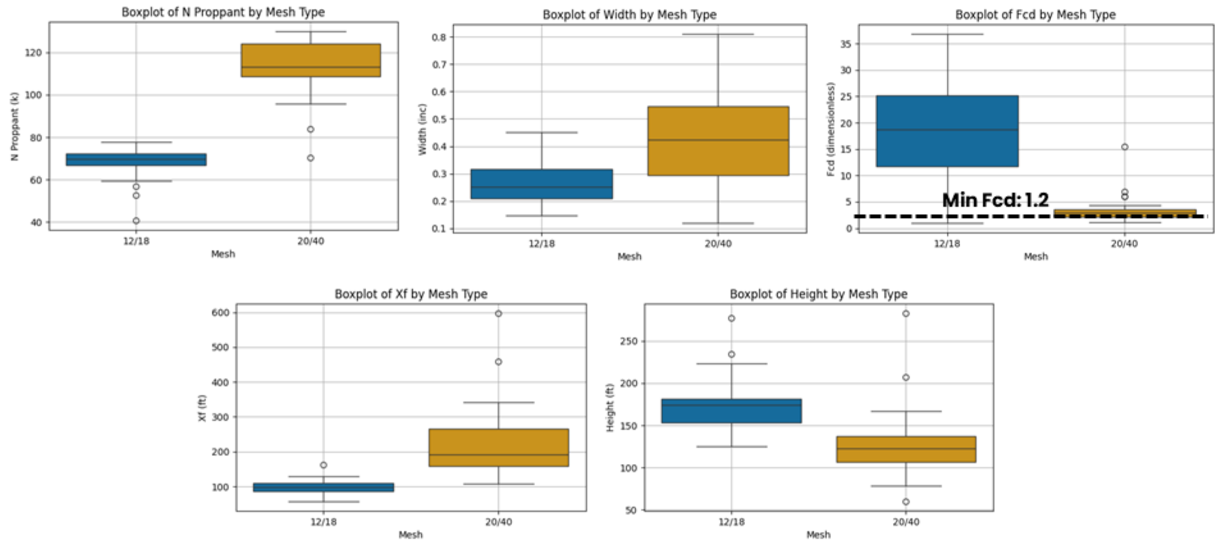


Figure 14—Fracturing Design Comparison

Outflow Behavior – The Role of Proppant Flowback

Further analysis reveals that the performance degradation is predominantly associated with outflow reliability. All wells using 20/40 mesh proppant experienced sand production, with 100% exhibiting significant proppant flowback [7]. This issue directly correlates with premature pump failure (Fig. 15), as shown by the shorter pump run life boxplot compared to 12/18 mesh wells. Poor pump reliability led to reduced fluid lifting, confirmed by normalized cumulative fluid production that fell below the operational target of 40 Mbbls in most 20/40 mesh cases.

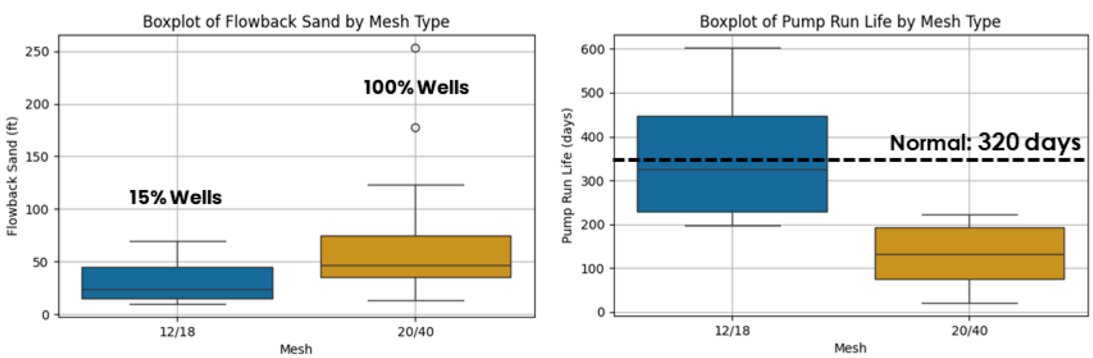


Figure 15—Pump Performance Analysis

A scatterplot of cumulative oil versus cumulative fluid further supports this diagnosis. The strong linear relationship between these two metrics demonstrates that poor fluid recovery directly limits oil production (Fig. 16). In other words, even with acceptable fracture geometry, if the fluid cannot be effectively lifted to surface due to pump failure or formation damage caused by sand fill, oil recovery will suffer.

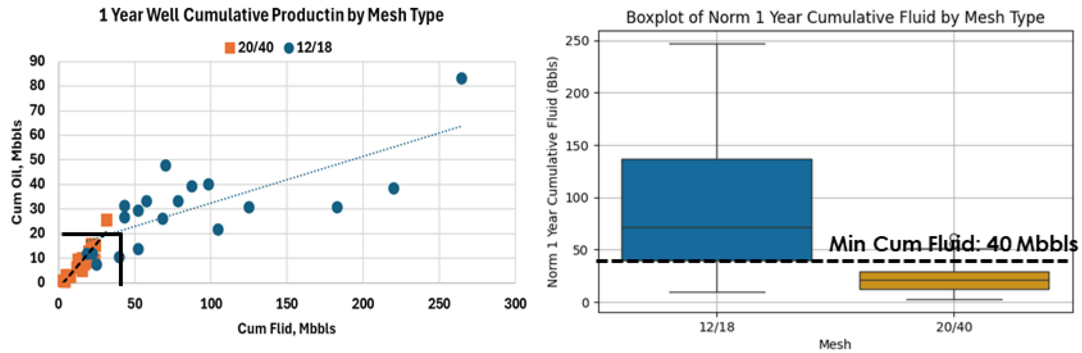


Figure 16—Production Performance

The integration of inflow normalization, fracture geometry evaluation, and outflow performance has revealed a consistent degradation pattern in 20/40 mesh wells, primarily driven by proppant flowback and associated mechanical failures. While the inflow analysis indicated no significant design flaws and acceptable conductivity ranges, the root cause remains incomplete without validating the integrity of the created fracture system. One hypothesis raised during early design revisions was the risk of proppant embedment, particularly in ductile formations like the TE reservoir. To investigate this, Pressure Transient Analysis (PTA) was performed to detect long-term changes in fracture conductivity and validate whether embedment occurred over time.

Proppant Embedment Validation through Pressure Transient Analysis

To further validate whether proppant embedment contributed to the performance of 12/18 mesh wells, a dedicated Pressure Transient Analysis (PTA) was conducted on five selected wells (Fig. 17) [8]. These wells had been fractured using 12/18 mesh and had produced for over 15 years, providing a robust dataset to evaluate long-term fracture conductivity retention.

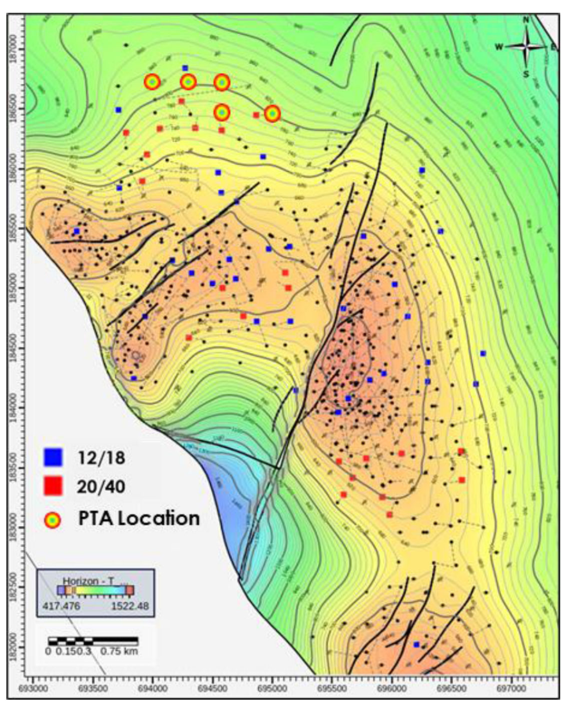


Figure 17—Wells PTA Location

The pressure transient analysis was conducted using a structured workflow (Figure 18) to ensure consistency, repeatability, and accuracy of results. The workflow began with acquiring test data, including pressure and flow rate datasets, complemented by PVT data and reservoir properties. A geological condition review was performed to provide contextual understanding of reservoir behavior, including layering, faulting, and lithofacies heterogeneity.

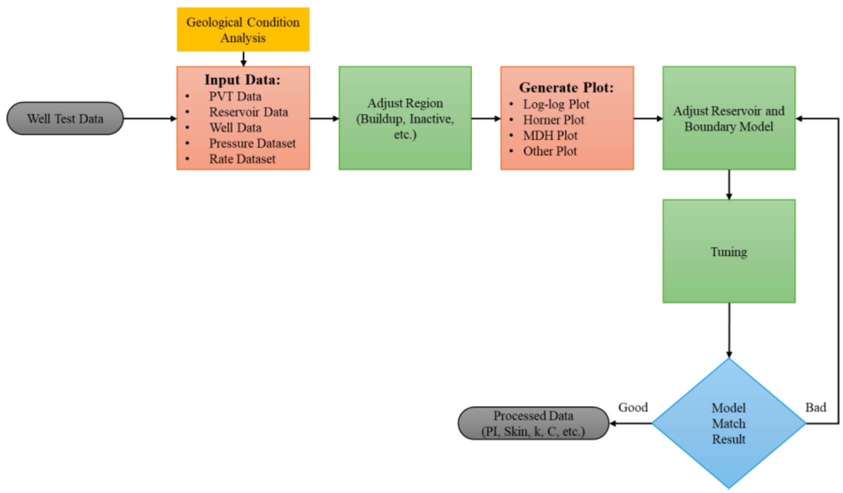


Figure 18—Pressure Transient Analysis Workflow

The input datasets were then processed, and flow periods were segmented to isolate buildup, drawdown, or inactive regions. Using this adjusted time-series dataset, various diagnostic plots such as log-log, Horner, and Miller-Dyes-Hutchinson (MDH) plots were generated for interpretation.

Based on the flow regimes observed in the plots, an initial reservoir and fracture model was developed and iteratively tuned using type-curve matching. Model outputs such as fracture half-length (Xf), and conductivity (Cf) were compared with post-frac job reports. The PTA results (Fig. 19) clearly demonstrate the presence of fracture-dominated flow, as evidenced by the linear behavior observed in the log-log derivative plots across all five wells. This indicates that the induced fractures remain effective in governing the flow regime, even after a decade of production.

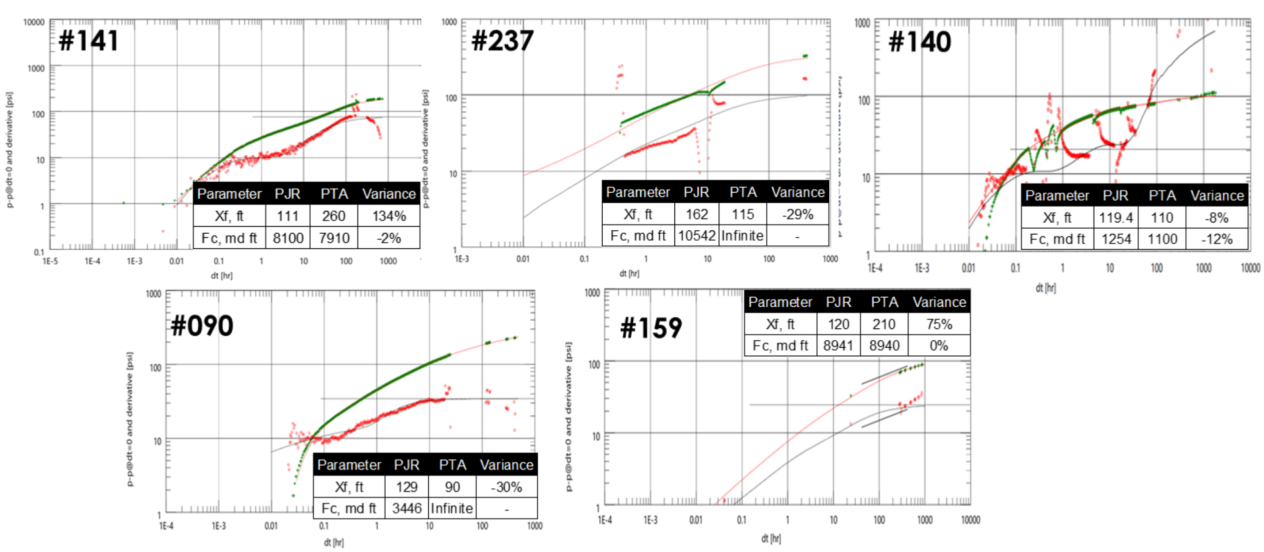


Figure 19—PTA Results

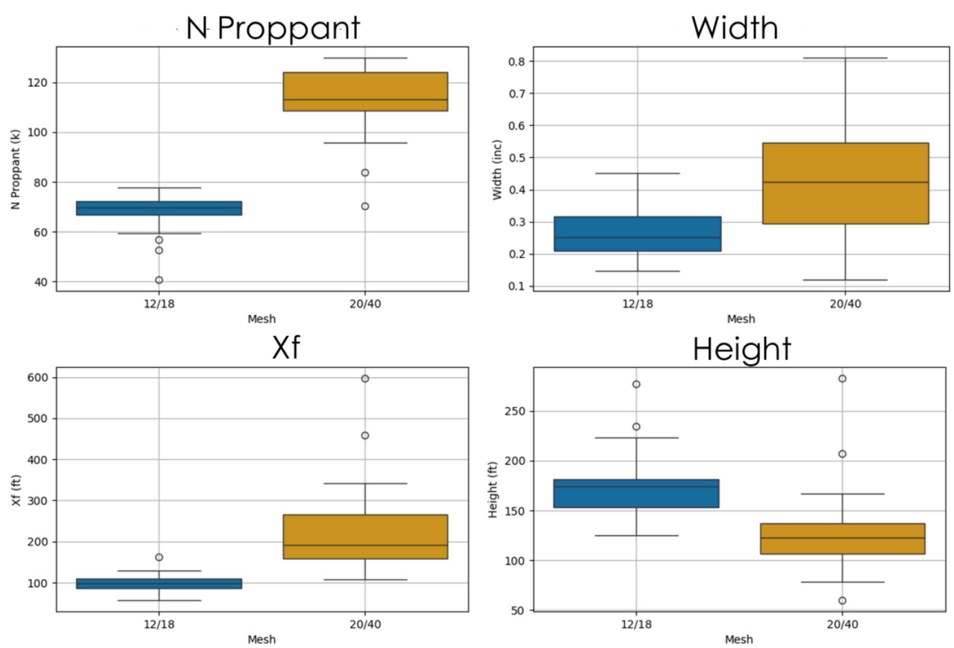
Furthermore, a comparison between the post-job report (PJR) and PTA-derived fracture geometry and conductivity values (Xf and Cf) revealed minimal variance across all wells. For example, well #141 shows only a 2% variance in Xf, while Well #159 displayed a 0% difference in conductivity. These findings confirm that the initial fracture geometry was largely preserved, and that no significant conductivity degradation occurred over time. As such, the feared risk of proppant embedment, typically expected in ductile formations with low closure pressure (~300 psi) was not substantiated by the field data.

The consistency in fracture performance despite the formation's low Young's modulus (~500 ksi) reinforces the technical viability of 12/18 mesh proppant in ductile, low-stress reservoirs. The results suggest that embedment is not a dominant mechanism affecting long-term conductivity, particularly when the stimulation is properly executed and supported by minimal proppant flowback.

These outcomes provide crucial validation that the use of larger mesh proppant sizes, such as 12/18, remains a sustainable choice for ensuring durable fracture conductivity in TE Formation-type reservoirs.

Optimized Fracturing Design

A comparative analysis of fracture geometry between 12/18 and 20/40 mesh designs was conducted to understand how proppant type influenced stimulation effectiveness. The comparison focused on key design outputs, namely proppant volume (klbs), fracture length, fracture width, and fracture height using both boxplots and summarized parameter ranges, as shown in Figure 20.



Parameter	12/18	20/40
N Proppant, klbs	66 - 72	108 – 124
Xf, ft	85 - 110	157 – 266
Width, inch	0.21 – 0.31	0.29 – 0.54
Height, ft	153 - 181	106 - 137

Figure 20—Fracturing Design Recommendation

The 20/40 mesh completions consistently required significantly higher proppant volumes, ranging from 108 to 124 klbs, compared to just 66 to 72 klbs in the 12/18 mesh completions. This increase in proppant volume correlated with longer fracture lengths (157–266 ft for 20/40 vs. 85–110 ft for 12/18) and wider widths (0.29–0.54 in for 20/40 vs. 0.21–0.31 in for 12/18). However, a trade-off was observed in the vertical propagation of the fracture: the 20/40 mesh designs resulted in shorter fracture heights, ranging between 106–137 ft, whereas the 12/18 designs achieved significantly taller fractures, from 153–181 ft.

This difference in fracture geometry had direct implications for stimulation effectiveness. Despite the theoretically higher conductivity expected from longer and wider fractures, the 20/40 mesh designs failed to sustain long-term performance. These wells exhibited higher proppant flowback, reduced cumulative production, and shorter pump run life, as previously discussed in the outflow section. On the other hand, the 12/18 mesh completions demonstrated more compact and vertically extended fracture geometries. The ability to propagate fractures higher within the reservoir zone contributed to improved drainage efficiency and better alignment with the reservoir's vertical heterogeneity. This, coupled with lower proppant requirements, resulted in a more efficient, cost-effective, and ultimately more productive stimulation strategy.

These findings highlight that optimal fracture geometry is not solely determined by fracture length or width but must consider the holistic stimulation response including reservoir contact, vertical sweep, and proppant placement sustainability. The 12/18 mesh clearly outperformed the 20/40 design in balancing these parameters, making it the preferred choice for future fracturing operations in the TE reservoir.

Pilot Success & Field Expansion Plan

To validate the field-scale applicability of the recommended fracturing strategy using 12/18 mesh proppant, two pilot wells (#541 and #561) were executed in November 2024 (Fig. 21). Both wells followed the optimized fracturing design, targeting improved fracture geometry and conductivity retention based on previous diagnostic findings.

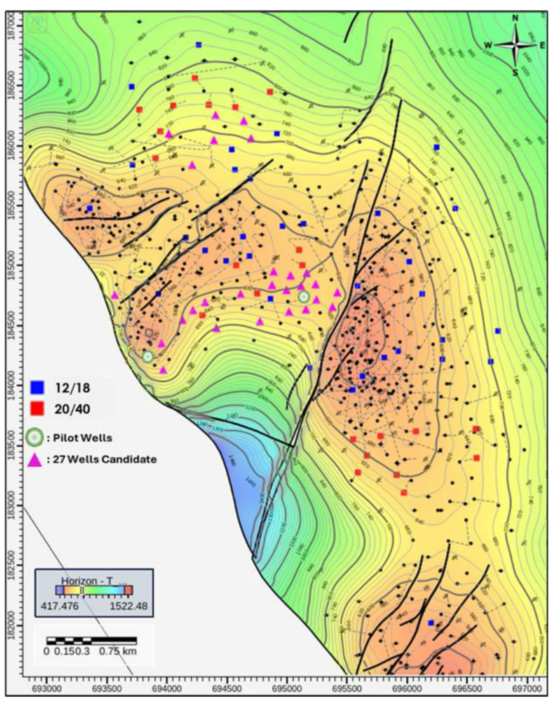


Figure 21—Pilot and Approved New Wells Location

Production results from the pilot wells demonstrated promising outcomes. As shown in Figure 22, both wells exceeded the One-year oil production plan and significantly outperformed historical wells completed using 20/40 mesh proppant. Notably, both pilot wells remained on production with no signs of early pump failure, which is commonly associated with proppant flowback [9]. This indicates that the 12/18 mesh design effectively maintained fracture conductivity and supported mechanical reliability in outflow systems.

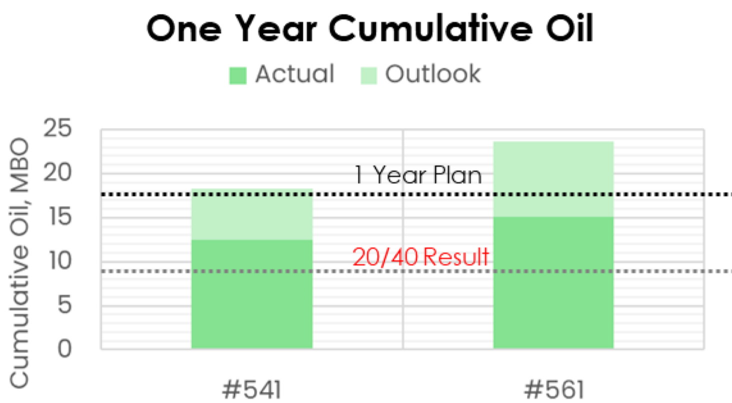


Figure 22—Pilot Wells Result

Economically, the pilot wells delivered a compelling outcome. Cumulative oil production increased by nearly 100% compared to historical performance of 20/40 mesh, while overall costs were reduced by 54% through lower proppant requirements and fewer well service interventions. Together, these factors contributed to a 127% improvement in Net Present Value (NPV), reinforcing the case for adopting the 12/18 mesh design in future full-field development [10].

Building on this economic and technical success, the pilot campaign provided strong justification for scaling up the optimized design. Consequently, a full-field development initiative has been approved, with 27 new wells sanctioned for drilling across the TE reservoir (Fig. 21). All wells will adopt the proven 12/18 mesh fracturing design to ensure consistency in performance and execution.

This transition to field-wide application represents a strategic step forward in unlocking the remaining oil potential of the TE reservoir. Given its classification as a low-quality and ductile formation, the adoption of a validated and effective stimulation design is expected to improve recovery, enhance economic returns, and establish a scalable model for future development across similar low-quality reservoirs in the Rokan Block.

Conclusion

A comprehensive analysis integrating inflow, outflow, and diagnostic evidence has led to several key conclusions regarding the performance disparity between wells completed using 12/18 and 20/40 mesh proppants. First, it was established that the performance differences are not driven by variations in reservoir quality or fracture geometry. Statistical testing confirmed that reservoir properties such as porosity, permeability, Young's modulus, and pressure gradient were consistent across both mesh types. Normalized productivity index (PI) analysis further revealed that wells using 12/18 mesh consistently outperformed those with 20/40 mesh over time.

The primary root cause of underperformance in 20/40 mesh wells was identified as proppant flowback, which led to premature pump failures and reduced cumulative oil and fluid production. This phenomenon compromised outflow reliability and contributed significantly to the observed PI decline. In contrast, 12/18 mesh completions showed minimal signs of flowback, longer pump run life, and more sustainable production outcomes.

Pressure Transient Analysis (PTA) conducted on five wells after 15 years of production validated that fracture conductivity and geometry remain intact, with minimal variance between post-job results and

PTA interpretations. The PTA results confirmed that fracture-dominated flow persists and that proppant embedment is not a significant concern, even in low-closure pressure and ductile formations. These findings support the continued use of larger mesh proppants such as 12/18 in similar environments.

Finally, this study also highlights an important lesson for reservoir and production engineers. Recommendations or design changes, even when suggested by experts, must be carefully validated against field data and thorough technical analysis. Applying engineering judgment in this way ensures that decisions remain aligned with the actual reservoir behavior and reduces the risk of adopting practices that may lead to suboptimal results.

Recommendations and Future Work

Recommendations

Based on the findings, it is strongly recommended to:

1. Continue using 12/18 mesh proppant for future fracturing operations to ensure better fracture conductivity retention and minimize proppant flowback risk.
2. Enhance pump reliability by implementing operational safeguards to reduce the risk of sand filling and mechanical failures.
3. Proceed with full field development, with 27 new wells already approved for drilling using the optimized 12/18 mesh fracturing design.
4. Maintain post-stimulation surveillance to validate the long-term effectiveness of the selected design, particularly in terms of pump run life and production sustainability.

Future Work

To extend the impact of this study, the following strategic steps are proposed:

1. Refine the fracturing design using feedback from ongoing field implementation and continuous economic evaluation.
2. Evaluate the scalability of the 12/18 mesh proppant strategy to other low-quality reservoirs across the Rokan Block, particularly those with similar ductile characteristics.
3. Leverage machine learning for predictive modelling to assess fracture design outcomes. This will support proactive decision-making and improve future stimulation design customization.

Acknowledgements

The authors sincerely thank PT Pertamina Hulu Rokan, particularly Balam Subsurface Development Team who support project team during the maturation and execution process. We also extend our appreciation to Pertamina's management and stakeholders for their support and permission in publishing this paper.

Nomenclature

Term	Definition
BSO	Field Name
Fcd	Dimensionless Fracture Conductivity
HQR	High-Quality Reservoir
LQR	Low-Quality Reservoir
MDH	Miller-Dyes-Hutchinson
NPV	Net Present Value
OHPL	Open-Hole Perforated Liner
PI	Productivity Index

PTA Pressure Transient Analysis
STB Stock Tank Barrels
Xf Fracture Half Length

References

1. Holditch, S.A. 2007. SPE Reservoir Stimulation, 3rd Edition. Richardson, TX: Society of Petroleum Engineers.
2. Economides, M.J., Nolte, K.G., Hill, A.D., and Economides, C.E. 2000. Reservoir Stimulation, 3rd Edition. New York: Wiley.
3. Poerwanto, J. H.; Sugembong, C. F.; Bagzis, J. M.; Martinez, A. D. 1995. "Application of Hydraulic Fracturing Technologies to the Shallow Telisa Formation." SPE-29282-MS. Presented at SPE Asia Pacific Oil and Gas Conference and Exhibition, Kuala Lumpur, Malaysia, 20–22 March. https://doi.org/10.2118/29282-MS
4. Barree, R.D., and Conway, M.W. 2002. "An Improved Method for Estimating Fracture Conductivity." SPE 77675. Presented at the SPE Annual Technical Conference and Exhibition, San Antonio, Texas, 29 September–2 October. https://doi.org/10.2118/77675-MS
5. Montgomery, D.C., and Runger, G.C. 2014. Applied Statistics and Probability for Engineers, 6th Edition. Hoboken, NJ: John Wiley & Sons.
6. Craft, B.C. and Hawkins, M., 1991. Applied Petroleum Reservoir Engineering. 2nd Edition. Englewood Cliffs, NJ: Prentice Hall.
7. Foster, J.T., et al 2004. "Proppant Flowback Prevention: Design and Field Results." SPE 90697. Presented at the SPE Annual Technical Conference and Exhibition, Houston, Texas, 26–29 September. https://doi.org/10.2118/90697-MS
8. Lee, J., and Wattenbarger, R.A. 1996. Gas Well Testing Handbook. Houston, TX: Gulf Publishing Company.
9. Internal Field Report, 2024. "Pilot Well Results, BSO Field." Operator Confidential Report (Unpublished).
10. Smith, J.L. 1990. Valuation of Oil and Gas Properties. Richardson, TX: Society of Petroleum Engineers.